Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2000-II-Q02] | Part A (Q1-Q5)

Count all integer-coordinate points (x, y) satisfying $x^2 - y^2 = 2000^2$.

求满足 $x^2 - y^2 = 2000^2$ 的整数点 (x, y) 个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2002-II-Q03] | Part A (Q1-Q5)

Increasing positive integers a, b, c form a geometric sequence, satisfy $\log_6 a + \log_6 b + \log_6 c = 6$, and have $b - a$ a perfect square. Find $a + b + c$.

递增正整数 a, b, c 构成等比数列，满足 $\log_6 a + \log_6 b + \log_6 c = 6$，且 $b - a$ 为完全平方数。求三数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-II-Q01] | Part A (Q1-Q5)

Three positive integers have product N equal to six times their sum, and one integer equals the sum of the other two. Add all possible distinct values of N.

三个正整数乘积为 N，且等于三数之和的六倍，其中一数又等于另两数之和。求 N 所有不同可能值之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2004-II-Q03] | Part A (Q1-Q5)

A rectangular block made of unit cubes is viewed with three adjacent faces visible. Exactly 231 unit cubes touch none of these faces. Find the least possible total number of cubes.

单位立方体拼成的长方体可见三个相邻面，恰有二百三十一个单元不接触任何可见面。求单元总数最小值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2005-I-Q04] | Part A (Q1-Q5)

A band has five members left over after making a square formation. It can instead form a complete rectangle with seven more rows than columns. Find the largest possible band size.

乐队排成正方阵后多出五人，但可排成行数比列数多七的完整长方阵。求最多人数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2008-I-Q03] | Part A (Q1-Q5)

Positive integer cycling, jogging and swimming rates b, j, s satisfy $2b + 3j + 4s = 74$ and $4b + 2j + 3s = 91$. Find $b^2 + j^2 + s^2$.

骑车、慢跑、游泳的正整数速度 b, j, s 满足 $2b + 3j + 4s = 74, 4b + 2j + 3s = 91$，求三者平方和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2008-I-Q04] | Part A (Q1-Q5)

Positive integers x, y satisfy $x^2 + 84x + 2008 = y^2$. Find $x + y$.

正整数 x, y 满足 $x^2 + 84x + 2008 = y^2$，求 $x + y$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2009-II-Q04] | Part A (Q1-Q5)

In an eating contest the first-place child eats n grapes, and place k eats $n+2-2k$, a nonnegative count. The total eaten is 2009. Minimize n.

吃葡萄比赛第一名吃 n 个，第 k 名吃 $n+2-2k$ 个且非负，总数为 2009。求最小 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2010-I-Q05] | Part A (Q1-Q5)

Positive integers obey $a > b > c > d$, $a + b + c + d = 2010$ and $a^2 - b^2 + c^2 - d^2 = 2010$. How many values of a are possible?

正整数满足 $a > b > c > d, a + b + c + d = 2010, a^2 - b^2 + c^2 - d^2 = 2010$。求 a 的可能值数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2012-II-Q01] | Part A (Q1-Q5)

Count positive integer pairs (m, n) solving $20m + 12n = 2012$.

求正整数解 (m, n) 的个数，使 $20m + 12n = 2012$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2021-I-Q05] | Part A (Q1-Q5)

Consider strictly increasing three-term arithmetic progressions of integers. Require that the sum of the three squared terms equals the middle term times the square of the common difference. Sum the last terms of all such progressions.

考虑严格递增的三项整数等差数列，要求三项平方和等于中项乘公差的平方。求所有此类数列末项之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2025-I-Q04] | Part A (Q1-Q5)

Count integer pairs $-100 \leq x, y \leq 100$ satisfying $12x^2 - xy - 6y^2 = 0$.

求满足 $-100 \leq x, y \leq 100$ 及 $12x^2 - xy - 6y^2 = 0$ 的整数有序对数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2026-I-Q04] | Part A (Q1-Q5)

How many integers at most 100 can be written as $a+b+ab$ with distinct positive integers a, b?

不超过 100 的整数中，有多少个可写成 $a+b+ab$，其中 a, b 是不同的正整数?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2000-I-Q06] | Part B (Q6-Q10)

Count integer pairs $0 < x < y < 10^6$ such that $(x+y)/2 = \sqrt{xy} + 2$.

求整数对 $0 < x < y < 10^6$ 的数量，使 $(x+y)/2 = \sqrt{xy} + 2$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2002-I-Q08] | Part B (Q6-Q10)

Among nondecreasing positive-integer sequences with $a_n = a_{n-1} + a_{n-2}$ for $n > 2$, find the least k that can be a_9 for two different sequences.

非降正整数数列满足 $a_n = a_{n-1} + a_{n-2}$，$n > 2$。求最小 k，使至少两个不同数列的第九项均为 k。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-I-Q07] | Part B (Q6-Q10)

On segment AC, point B satisfies $AB = 9, BC = 21$. A point D off line AC has equal integer lengths $AD = CD$ and integer BD. Sum all possible perimeters of triangle ACD.

线段 AC 上点 B 满足 $AB = 9, BC = 21$。点 D 不在直线 AC 上，且 $AD = CD$ 为整数、BD 也为整数。求 ACD 所有可能周长之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-I-Q08] | Part B (Q6-Q10)

Four increasing positive integers have their first three terms arithmetic and their last three terms geometric. The fourth exceeds the first by 30. Find their sum.

四个递增正整数的前三项等差、后三项等比，第四项比第一项大 30。求四数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-II-Q10] | Part B (Q6-Q10)

Two positive integers differ by 60, and the sum of their square roots equals the square root of a nonsquare integer. Maximize their sum.

两正整数差为 60，其平方根之和等于某非完全平方整数的平方根，求两整数和的最大值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2004-II-Q06] | Part B (Q6-Q10)

Three monkeys successively take portions of a banana pile. The first keeps $3/4$ of its portion and splits the rest between the others; the second keeps $1/4$; the third takes the remainder and keeps $1/12$, again splitting the rest equally. Every received amount is an integer, and final totals have ratio $3:2:1$. Find the least initial pile size.

三猴依次取香蕉。第一猴留所取的四分之三，余数平分给另两猴；第二猴留四分之一并平分余数；第三猴取全部剩余，留十二分之一并平分余数。每次所得均为整数，最后总数比为 $3:2:1$。求最小初始总数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2009-II-Q09] | Part B (Q6-Q10)

Let m, n count positive integer solutions of $4x + 3y + 2z = 2009$ and $4x + 3y + 2z = 2000$, respectively. Find $(m - n) \bmod 1000$.

令 m, n 分别为 $4x + 3y + 2z = 2009$ 和 $4x + 3y + 2z = 2000$ 的正整数解数，求差除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2013-II-Q06] | Part B (Q6-Q10)

Find the smallest positive N such that the thousand consecutive integers $1000N, \ldots, 1000N + 999$ contain no perfect square.

求最小正整数 N，使从 $1000N$ 至 $1000N + 999$ 的一千个连续整数中没有完全平方数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2013-II-Q07] | Part B (Q6-Q10)

Clerks sort 1775 files at thirty files per clerk-hour. Equal numbers are reassigned after hours one, two, and three, and the work finishes after three hours ten minutes. How many files were sorted during the first ninety minutes?

若干职员每人每小时整理三十份文件，共整理一千七百七十五份。在第一、二、三小时末各调走相同人数，三小时十分完成。求前九十分钟整理份数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2015-II-Q08] | Part B (Q6-Q10)

Positive integers a, b satisfy $(ab+1)/(a+b) < 3/2$. Write the maximum of $(a^3b^3+1)/(a^3+b^3)$ as reduced p/q and find $p+q$.

正整数 a, b 满足 $(ab+1)/(a+b) < 3/2$。将 $(a^3b^3+1)/(a^3+b^3)$ 的最大值写成最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2017-II-Q06] | Part B (Q6-Q10)

Find the sum of positive integers n for which $n^2 + 85n + 2017$ is an integer square.

求使 $n^2 + 85n + 2017$ 为整数平方的所有正整数 n 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2004-II-Q14] | Part C (Q11-Q15)

For how many positive integers n can plus signs inserted between a string of n sevens give sum 7000?

对于多少个正整数 n，在连续 n 个七组成的数位串之间插加号后能得到和七千?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2005-II-Q13] | Part C (Q11-Q15)

An integer-coefficient polynomial satisfies $P(17)=10, P(24)=17$. If $P(n)=n+3$ has two distinct integer solutions n_1, n_2, find $n_1 n_2$.

整系数多项式满足 $P(17)=10, P(24)=17$。若 $P(n)=n+3$ 有两个不同整数解 n_1, n_2，求其乘积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2008-II-Q15] | Part C (Q11-Q15)

Find the largest integer n whose square is a difference of consecutive integer cubes and for which $2n + 79$ is square.

求最大整数 n，使 n^2 为两个连续整数立方之差，且 $2n + 79$ 为平方数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2009-I-Q14] | Part C (Q11-Q15)

For 350 entries $a_i \in \{1,2,3,4\}$, let $S_t = \sum_i a_i^t$. Given $S_1 = 513, S_4 = 4745$, minimize S_2.

三百五十个数 $a_i \in \{1,2,3,4\}$，令 $S_t = \sum_i a_i^t$。已知 $S_1 = 513, S_4 = 4745$，求 S_2 最小值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2011-I-Q15] | Part C (Q11-Q15)

For some integer m, the polynomial $x^3 - 2011x + m$ has three integer roots a, b, c. Find $|a| + |b| + |c|$.

某整数 m 使 $x^3 - 2011x + m$ 有三个整数根 a, b, c。求 $|a| + |b| + |c|$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2017-I-Q13] | Part C (Q11-Q15)

For integer $m \geq 2$, let $Q(m)$ be the smallest positive integer such that every integer $n \geq Q(m)$ admits an integer cube k^3 satisfying $n < k^3 \leq mn$. Find $\sum_{m=2}^{2017} Q(m)$ modulo1000.

对整数 $m \geq 2$，令 $Q(m)$ 为最小正整数，使每个整数 $n \geq Q(m)$ 都有整数立方 k^3 满足 $n < k^3 \leq mn$。求 $\sum_{m=2}^{2017} Q(m)$ 模1000的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程